

Daniel J Noh — (he/him)

Ph.D. Student, University of Pennsylvania

dnoh@upenn.edu

danieljnoh.com | [Google Scholar](https://scholar.google.com/citations?user=...)

Research Interests

learning sciences, human-computer interaction, care & maintenance, design & computing education, algorithmic justice, constructionism, participatory design, design of learning spaces, accessibility

Education

Expected 2029	University of Pennsylvania <i>Ph.D.</i> , Learning Sciences & Technologies Advisor: Yasmin B. Kafai
2022	Harvard University <i>Ed.M.</i> , Learning Design, Innovation & Technology
2021	Carnegie Mellon University <i>B.Arch.</i> , Architectural & Building Sciences/Technology (University & College Honors) Minors: Human-Computer Interaction & Design for Learning

Research Experience

Aug 2024 to current	University of Pennsylvania Graduate School of Education — Kafai Lab (PI: Yasmin Kafai) NSF Grants #2333469, #2342438, #2414590 Spencer, Kapor, William T. Grant, Alfred P. Sloan Foundations Rapid Response Funding Google Academic Research Award: AI for Privacy, Safety, & Security (2025) <i>Research Fellow</i> - Collecting and analyzing data to investigate learners' and educators' conceptions of AI/ML systems—including generative language models, social media, and advertisements—and algorithmic justice in formal and informal learning environments - Developing related learning tools and activities through participatory design with educators and youth
Jun 2024 to Aug 2024	Northeastern University College of Arts, Media and Design (PI: Sara Hendren) <i>Research Assistant</i> Developed experimental pedagogical modules for the studio-seminar course, <i>Investigating Normal: Design and Disability</i>, identifying best practices for accessible spatial and digital design, related precedent projects, and assessment strategies for proposed activities and curricular prompts
Sep 2021 to Jun 2022	Harvard University Project Zero — Designing Learning Places Lab (PI: Daniel Wilson) <i>Research Assistant</i> - Conducted a literature review (white paper) on the qualities of materials, objects, and environments that support the learning practices of noticing, wondering, and help-seeking - Designed a practical toolkit for practitioners to adopt the findings into the design of learning spaces - Co-developed the course, <i>Designing Learning Places</i>, currently taught at the Harvard Graduate School of Education

Sep 2020 to
Sep 2021 | **Carnegie Mellon University**
Human-Computer Interaction Institute — ClassInSight (PI: Amy Ogan)

James S. McDonnell Foundation Grant #1822813

Research Associate (May 2021 to Sep 2021)

Research Assistant (Sep 2020 to May 2021)

- Drafted and conducted focus group protocols, interview questions, and participatory design sessions to understand how classroom discussion data visualizations and personalized professional development can scaffold teacher agency and meaning-making
- Designed prototypes and data visualizations for a conversation support professional development tool

Honors & Awards

2024-2028 | **Fontaine Society Fellowship**, University of Pennsylvania
2021 | **GEE! Learning Games Award Finalist**, GEE! Learning Games Awards
2020 | **Askwith Kenner Artist Grant**, CMU Department of Modern Languages
2020 | **Louis F. Valentour Scholarship**, CMU School of Architecture

Publications

Journal Articles (Peer-Reviewed)

- [J.1] | Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. B. (2025). High school students building babyGPTs: Engaging in data practices and addressing ethical issues through the construction of generative language models. *International Journal of Child-Computer Interaction*, 45, 100769.
DOI: <https://doi.org/10.1016/j.ijcci.2025.100769>

Conference Publications (Peer-Reviewed, Full)

- [C.4] | Morales-Navarro, L., **Noh, D. J.**, Servat, L., Netting, C., Kafai, Y. B., & Metaxa, D. (2026, Accepted). Building to Understand: Examining Teens' Technical and Socio-Ethical Pieces of Understanding in the Construction of Small Generative Language Models. *25th Interaction Design and Children*.
- [C.3] | **Noh, D. J.**, Fields, D. A., Kafai, Y. B., & Metaxa, D. (2026, Accepted). "You Can Actually Do Something": Shifts in High School Computer Science Teachers' Conceptions of AI/ML Systems and Algorithmic Justice. *International Conference of the Learning Sciences*.
- [C.2] | **Noh, D. J.** (2026). Guide on the Side or Sage on the Stage?: Exploring the Relationship Between Teachers' Spatial and Verbal Discursive Strategies. In: G. Carmona, C. Lima, M. J. Santos, H. Benítez, L. Montero-Moguel, & B. Galarza-Tohen (Eds.), *Advances in Quantitative Ethnography* (Vol. 2677, pp. 509–523). Springer Nature Switzerland.
DOI: https://doi.org/10.1007/978-3-032-12229-2_33
- [C.1] | **Noh, D. J.**, Fields, D. A., Morales-Navarro, L., Cabrera-Sutch, A. M., Kafai, Y. B., & Metaxa, D. (2025). Youth as Advisors in Participatory Design: Situating Teens' Expertise in Everyday Algorithm Auditing with Teachers and Researchers. *Proceedings of the 24th Interaction Design and Children*, 415–428.
DOI: <https://doi.org/10.1145/3713043.3728849> [Acceptance Rate: 29%]

Conference Publications (Peer-Reviewed, Short & Symposia)

- [S.4] | Morales-Navarro, L., Kafai, Y., Hilke, O., Pope, N., Tedre, M., **Noh, D. J.**, Fields, D. A., Metaxa, D., Solyst, J., Shapiro, R. B., Fong, C., Nurdin, F., Sharma, S., Iivari, N., Klemettilä, P. A., Bilstrup, K-E., Musaeus, L. H., Landesman, R., Everson, J., Ryoo, J. & Iversen, O. S. (2026, Accepted). Learning to Deconstruct AI/ML Systems: Scaffolds to Support Learners in Critically Evaluating AI/ML Systems. *International Conference of the Learning Sciences*.

- [S.3] Ching, C., Cortez, A., Higgs, J., Grover, S., Vakil, S., Paik, W., Abdul, S., Knight, M., Ryoo, J., Choi, M., Wei, W., Clarke, R., Blizzard-Caron, J., Stark, L., Kohn, L., Voloch, D., Ochterski, J., Lizárraga, J., Wang, X. C., Xing, G., Dantu, K., Morales-Navarro, L., **Noh, D.**, Kafai, Y. & Stornaiuolo, A. (2026, Accepted). Artificial Intelligence and Human Purposes: Dialogues about Creativity and Control. *International Conference of the Learning Sciences*.
- [S.2] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. (2025). Building babyGPTs: Youth engaging in data practices and ethical considerations through the construction of generative language models. *Proceedings of the 24th Interaction Design and Children*, 1021–1026.
DOI: <https://doi.org/10.1145/3713043.3731525>
- [S.1] Kafai, Y., Shapiro, R. B., Jetzinger, F., Michaeli, T., Tedre, M., Vartiainen, H., Iivari, N., Musaeus, L. H., Iversen, O. S., Ali, S., Bodon, H., Butler, M., Kshirsagar, K., Smith, M., Quiterio, A., Worsley, M., Kumar, V., Morales-Navarro, L., **Noh, D.**, Pea, R., & Philip, T. M. (2025). Youth as Designers of Artificial Intelligence and Machine Learning Technologies: What Do We Know About the Opportunities and Challenges of K-12 Students Creating Their Own Applications?. In *Proceedings of the 19th International Conference of the Learning Sciences*, 2260-2268. International Society of the Learning Sciences.
DOI: <https://doi.org/10.22318/icls2025.362276>

Posters, Presentations & Workshops (Peer-Reviewed)

- [P.7] Porter, B., Okker-Edging, K., Liu, A., **Noh, D. J.**, Lydic, K., Muzekari, B., Lewis, T., Torres-Grillo, O., Falk, E., Tan, A. S.L. & Cosme, D. (2026, May). Youth Participatory Action Communication Research on Social and Environmental Determinants of Health. *2026 Conference for Advancing Participatory Sciences*. Virtual.
- [P.6] **Noh, D.** Fields, D. A., Kafai, Y. B. & Metaxa, D. (2026, April). “A Path to Activism”: CS Teachers Developing Critical Agency with Algorithmic Systems through Algorithm Auditing. *2026 American Educational Research Association (AERA) Conference*. Los Angeles, CA.
- [P.5] Morales-Navarro, L., **Noh, D.** & Kafai, Y. B. (2026, April). High School Students as Designers of Generative Language Models. *2026 American Educational Research Association (AERA) Conference*. Los Angeles, CA.
- [P.4] Morales-Navarro, L., **Noh, D.** & Kafai, Y. B. (2026, April). Teens’ Ethical Considerations When Building babyGPTs. *2026 American Educational Research Association (AERA) Conference*. Los Angeles, CA.
- [P.3] Fields, D. A., Morales-Navarro, L., **Noh, D.**, Kafai, Y. B. & Ottina, J. (2025, October). Workshop: Investigating the Black Box of Artificial Intelligence and Machine Learning: Promoting Youth and Teacher Agency through Algorithm Auditing. *Connected Learning Summit 2025*. Virtual.
- [P.2] **Noh, D. J.**, Morales-Navarro, L., & Kafai, Y. (2025, October). What Comes Next?: Youth Learning About and Designing Markov Chains through Unplugged Activities. *Connected Learning Summit 2025*. Virtual.
- [P.1] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. (2025, October). What happens when teens design small generative language models? A case study of teenagers building babyGPTs. *Connected Learning Summit 2025*. Virtual.

Public Scholarship (White Papers, Curriculum & Other)

- [PS.2] Butapetch, H., Fields, D., Hanna, S., Landa, J., Morales-Navarro, L., Myers, T., **Noh, D.**, Ottina, J., & Ulrich, A., with Kafai, Y. & Metaxa, D. (2025). Introduction to Algorithm Auditing: Lessons for High School Computer Science (beta version).
Available at: <https://bit.ly/k12aiaudit-lessons>
- [PS.1] Gonzalez, P. G., **Noh, D.**, & Wilson, D. (2022). Making the Space for Learning. *Presidents and Fellows of Harvard College*. Cambridge, MA.
Available at: <https://pz.harvard.edu/resources/making-the-space-for-learning>

Teaching

Jan 2022 to Jan 2025	Harvard University Graduate School of Education <i>*As a Teaching Fellow I was involved in course redesign, development of teaching material, leading weekly in-class lectures/workshops, and providing constructive student feedback</i> <i>Teaching Fellow, Rapid Prototyping of Educational Products</i> Instructor: Bertrand Schneider (Winter 2025) <i>Lead Teaching Fellow & Course Co-Designer, Designing Learning Places</i> Instructor: Daniel Wilson (Spring 2023, Spring 2024) <i>Teaching Fellow, Digital Fabrication and Making in Education</i> Instructor: Bertrand Schneider (Spring 2022, Fall 2022, Spring 2024) <i>Teaching Fellow, Transforming Education Through Emerging Technologies</i> Instructor: Bertrand Schneider (Spring 2024) <i>Teaching Fellow, Multimodal Learning Analytics</i> Instructor: Bertrand Schneider (Spring 2023)
Feb 2024 to May 2024	Design Museum Everywhere / CoDesign Collaborative High School Inclusive Design Challenge (Co-Instructor: Rosa Weinberg) <i>Design Educator</i> Planned and facilitated a three-week long program on accessibility, inclusive design, and participatory design with six financially-compensated high school students
Sep 2022 to Jan 2024	Massachusetts Institute of Technology MIT Museum <i>Technical Instructor / Museum Educator</i> - Designed and facilitated novel hands-on STEAM learning experiences and public events to build meaningful connections to scientific inquiry and design exploration - Established standard operating procedures and trained museum educators on various fabrication tools in the makerspace (laser cutter, 3D printers, and vinyl cutters) - Mentored MIT undergraduate research assistants (UROP) and museum teen interns on the development and facilitation of design-based, hands-on learning experiences - Co-led the launch and development of the Design Redefined series, co-hosted by MIT MAD and Innovators for Purpose

Advising & Mentoring

UG = Undergraduate

Current	University of Pennsylvania - Lucianne Servat (UG, Cognitive Science, 2025-2026, with Luis Morales-Navarro) - Elo Esalomi (UG, Artificial Intelligence, 2025, with Luis Morales-Navarro)
2023-2024	Massachusetts Institute of Technology (Museum + MAD) - Wonuola Abiodun (UG, Civil Engineering + Architecture, 2023-2024, with Rosa Weinberg) - Alexandra Coston (UG, Architecture, 2023-2024, with Rosa Weinberg) - Kimberly McPherson (UG, Computer Science, 2023, with Rosa Weinberg)

Service

Journal Reviewing	Journal of the Learning Sciences (2026)
Conference Reviewing	ACM Interaction Design & Children (2026) International Conference of the Learning Sciences (2026) Educational Advances in Artificial Intelligence (2026) Connected Learning Summit (2025) International Conference on Quantitative Ethnography (2025)
Institutional Service	Massachusetts Institute of Technology - MIT-Nord Anglia, STEAM Teacher Professional Development Mentor (2024) Carnegie Mellon University - School of Architecture, Student Portfolio Reviewer (2021) - School of Architecture, Carnival Pavilion Build Team (2019) - School of Architecture, Architecture Peer Mentor (2017)
Other Service	Carnegie Museum of Natural History Museum Volunteer (2020)

Other Experience

Aug 2021 to May 2022	Harvard University Graduate School of Education – Education Innovation Studio <i>Makerspace Assistant</i> Taught graduate students how to use fabrication tools including laser cutters, 3D printers, vinyl cutters, sewing machine, and power tools
May 2020 to Aug 2020	Carnegie Museum of Natural History Center for Anthropocene Studies (Advisors: Asia Ward & Marti Louw) <i>Center for Anthropocene Studies Education Intern</i> - Produced educational videos about sewage treatment and potable water, presented at ALCOSAN's 2020 and 2021 Open House - Published corresponding articles in the museum's monthly newsletters and website
May 2018 to Jan 2021	Carnegie Mellon University School of Architecture <i>Design Fabrication (dFab) Lab Assistant</i> Assisted architecture students and faculty on implementing digital fabrication methods (laser cutters, 3D printers, CNC-mill, and vacuum formers) into their project workflow

Skills & Tools

Programming	Python, JavaScript, HTML/CSS, R
Fabrication	Laser Cutting, 3D Printing, Vinyl Cutting, Vacuum Forming, Sewing, Woodworking, Soldering, Basic Electronics/Microcontrollers
Design	Rapid Prototyping, Wireframing, Storyboarding, Illustrating, Data Visualization, Graphic Design, Architectural Design, Learning Design, Video Editing, Web Design & Development
Methods	Thematic Analysis, Interviews, Participatory Design Research, Youth Participatory Action Research, Multi-Modal Learning Analytics, Epistemic/Ordered Network Analysis

Professional Memberships

American Education Research Association (AERA)
International Society of the Learning Sciences (ISLS) – DIV C
Association of Computer Machinery (ACM) – SIGCHI